Nicolás Castro-Perdomo

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Address: 1001 E 10th St, 47405 Bloomington, Indiana, USA.

Education

Indiana University — Bloomington, United States
Ph.D. in Geological Sciences — *August 2022 - Present*
GPA: 3.9/4.0

Indiana University — Bloomington, United States
M.Sc. in Geological Sciences — *March, 2026*
GPA: 3.9/4.0

King Abdullah University of Science and Technology — Thuwal, Saudi Arabia
M.Sc. in Earth Science and Engineering — *December, 2019*
GPA: 3.85/4.0

Universidad Nacional de Colombia — Bogotá, Colombia
B.Sc. in Geology · Honors Degree — *September, 2017*
GPA: 4.4/5.0

Research Experience

Graduate Student Researcher — Bloomington, USA
Indiana University — *August, 2022 - Present*
Advisor: Kaj Johnson

Visiting Research Intern — Thuwal, Saudi Arabia
King Abdullah University of Science and Technology — *March, 2022 - August 2022*
Advisor: Sigurjón Jónsson

Graduate Student Researcher — Thuwal, Saudi Arabia
King Abdullah University of Science and Technology — *August, 2017 - December, 2019*
Advisor: Sigurjón Jónsson

Visiting Research Intern — Strasbourg, France
EOST, Université de Strasbourg — *June, 2018 - July, 2018*
Advisor: Frédéric Masson

Visiting Research Intern — Ithaca, United States
Cornell University — *January, 2016 - July, 2016*
Advisor: Larry D. Brown

Publications

1. **Nicolás Castro-Perdomo**, Sigurjón Jónsson, Yann Klinger, Frederic Masson, Thorsten Wolfgang Becker, and Kaj Johnson, **2025**. Strain rates along the Alpine-Himalayan belt from a comprehensive GNSS velocity field. *JGR: Solid Earth*. DOI: 10.1029/2025JB031738
2. Xing Li, Sigurjón Jónsson, Shaozhuo Liu, and Zhangfeng Ma, **Nicolás Castro-Perdomo**, Simone Cesca, Frédéric Masson, and Yann Klinger, **2024**. Resolving the slip-rate inconsistency of the northern Dead Sea fault. *Science Advances*. DOI: 10.1126/sciadv.adj8408
3. **Nicolás Castro-Perdomo**, Renier Viltres, Frédéric Masson, and Yann Klinger, Shaozhuo Liu, Maher Dhahry, Patrice Ulrich, Jean-Daniel Bernard, Rémi Matrau, Abdulaziz Alothman, Hani Zahran, Robert Reilinger, P Martin Mai, and Sigurjón Jónsson, **2022**. Interseismic Deformation in the Gulf of Aqaba from GPS Measurements. *Geophysical Journal International*. DOI: 10.1093/gji/ggab353

Manuscripts in preparation

4. **Nicolás Castro-Perdomo** and Kaj Johnson, **in prep**. Satellite Geodesy Reveals Substantial Aseismic Off-fault Deformation in California.
5. **Nicolás Castro-Perdomo** and Kaj Johnson, **in prep**. Fault Complexity Controls Deformation Partitioning Between Faults and the Surrounding Crust.

Software

1. **Nicolás Castro-Perdomo (2024)**. FICORO_GNSS: An open-source Python software package for filtering, combining and rotating GNSS velocity fields. DOI: 10.5281/zenodo.13921188.

Honors and Awards

College of Arts and Sciences Dissertation Research Fellowship — Bloomington, USA
Indiana University — *April, 2026*

NSF travel award (administered through Stanford University) — Smolenice, Slovakia
Numerical Modeling of Earthquake Motions Workshop — *June, 2024*

Graduation with honors (highest distinction) — Bogotá, Colombia
Universidad Nacional de Colombia — *September, 2017*

Valedictorian - Class rank: 1/40 — Bogotá, Colombia
Universidad Nacional de Colombia — *September, 2017*

First place award — Thuwal, Saudi Arabia
6th KAUST Annual International Undergraduate Research Poster Competition — *January, 2017*

Nexo Global - Colciencias scholarship — Ithaca, USA
Fully funded research internship at Cornell University — *January-July, 2016*

Dean's list - Tuition fee waiver — Bogotá, Colombia
Universidad Nacional de Colombia — *2013 - 2017*

Invited Presentations and Keynotes

European Geosciences Union (EGU) General Assembly 2026 — Vienna, Austria
"On-fault and off-fault deformation partitioning throughout the earthquake cycle" — *May, 2026*

Earth Sciences New Zealand (Formerly GNS Science) — Online
"Slip Deficit Rate Inversions in New Zealand from Geodetic Data" — *October, 2025*

COMET webinar series — Online
"A Comprehensive Model of Crustal Deformation in the Mediterranean" — *November, 2023*

Summer Courses

International School of Physics "Enrico Fermi" — Varenna, Italy
Italian Physical Society — *July 2018 (1 week)*
Course 202: Mechanics of Earthquake Faulting

Workshop on Mechanics of the Earthquake Cycle — Trieste, Italy
International Centre for Theoretical Physics (ICTP) — *October 2023 (2 weeks)*
Course SMR-3885

Workshop on Numerical Modeling of Earthquake Motions — Smolenice, Slovakia
Slovak Academy of Sciences — *June 2024 (1 week)*
NMEM 2024: Waves and Ruptures

Volunteer and Professional Service

Student Representative

American Geophysical Union (AGU). Geodesy section, 2025-2027

Peer Reviewer

JGR: Solid Earth, Tectonophysics, GJI

Skills

Computer Skills

Programming: Python, MATLAB, Bash, Julia (learning)
GNSS data analysis: GAMIT/GLOBK, CATS, HECTOR
Others: HPC-SLURM, PSGRN/PSCMP, TDEFNODE, GMT, LaTeX
OS: GNU/Linux, MacOS, Windows

Language Skills

Spanish: First language
English: C1 - IELTS 7.0

Fieldwork Experience

Geodetic survey — Gulf of Aqaba, Saudi Arabia
Surveyed 30 campaign GNSS stations — *March-April 2022 (2 weeks)*

Drone survey — Húsavik, Iceland
Collaborated in the acquisition of drone imagery — *July 2018 (1 week)*

CorNET seismic network — Ithaca, United States
Collected data from 3 seismic stations — *June, 2016 (2 days)*